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Context-Grounded Personalised Question-Answering under Centralised and Federated Training

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The work contained in this thesis is my own work unless otherwise stated.

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Abstract

Large language models are increasingly used as personalised assistants. Teaching them to base an answer on what a user has said in earlier conversations normally means collecting many users' histories in one place, which raises privacy concerns. This thesis studies how to train such a model in both settings: centrally, where the data is pooled, and federated, where each user's history stays where it is. Given a query and a relevant conversation snippet, we compare five training objectives: supervised fine-tuning, an answer-contrastive and a context-contrastive objective, a combination of the two, and a draft-revision objective that rewrites a general, non-personalised draft into the preferred answer. On PersonaMem-v2, draft-revision gives the highest personalisation success, as judged by a frontier language model, in both settings, while the individual contrastive objectives do not clearly beat supervised fine-tuning. Federated training lowers the score of every method, but the draft-revision advantage remains. This suggests that learning to correct a general answer is a stronger and more robust personalisation signal than learning to rank answers by preference.

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1 Introduction

We begin with personalised language models, and with why conversational settings make personalisation both natural and difficult. We then turn to the privacy cost of learning from conversation histories, and to federated training as a response to it. Next we separate personalisation from explicit preferences, which is comparatively well studied, from the implicit reasoning this thesis is about, and review the retrieval, memory, and learning signals available for it. The section closes with the scope of the work, its research question, the training objectives we compare, and our contributions.

1.1 Personalised Language Models

Large language models (LLMs) are increasingly used as interactive assistants for writing, planning, recommendation, education, and decision support (Liu et al., 2025; Wang et al., 2024b). As these systems move from one-off query answering into sustained dialogue, users increasingly expect them not only to be correct and helpful, but also to adapt to individual preferences, circumstances, and needs (Zhang et al., 2024b). A response that is appropriate for one user may be unsuitable for another: people differ in interests, habits, expertise, communication style, and personal constraints (Salemi et al., 2024b; Zhao et al., 2025). This has motivated growing interest in *personalised language models*, systems that adapt their behaviour using information about the individual user (Mysore et al., 2024; Zhang et al., 2024b).

Personalisation is especially natural in conversational settings, where such user-specific information accumulates over time. Over repeated interactions, an assistant can observe routines, constraints, prior knowledge, and stylistic preferences that make later answers more useful. For example, a recommendation should avoid suggestions that conflict with known user preferences, and an educational assistant may adjust the depth of an explanation to the user’s background. In this sense, personalisation shifts language models away from producing a single generally acceptable answer towards generating responses that are appropriate for a particular person.

1.2 Privacy and the Case for Federated Training

Personalisation of this kind depends on what a user has revealed about themselves over time, and that information is often sensitive. Conversation histories may reveal daily activities, relationships, health, workplaces, locations, and other private details. Training personalised models by pooling these histories on a central server therefore raises privacy, security, and data-governance concerns. These issues are becoming more pressing as smaller language models become suitable for deployment on personal devices (Wang et al., 2024a; Xu et al., 2024; Zhang et al., 2025a), and in domains such as healthcare and enterprise software, where confidential data naturally remain distributed

across devices, departments, or organisations. Local deployment, however, creates a learning problem: a model that sees only one user’s interactions cannot benefit from patterns shared across many users unless some form of collaboration is possible without collecting raw histories centrally.

Federated learning (McMahan et al., 2017) provides a natural framework for this setting. Clients keep private interaction data local, communicate gradients (or other model updates) that are aggregated into a shared model, and never upload raw histories. Federation alone does not guarantee complete privacy: information can still leak through updates, but it substantially reduces the need to centralise raw user data. The broader motivation of this thesis lies in that tension: the data a model needs in order to personalise is exactly the data that should not leave the user’s device. The question is whether useful personalised capabilities can still be learned under these constraints.

1.3 Explicit Preferences and Implicit Personalised Reasoning

Answering that question first requires being precise about what kind of personalisation is at stake, because the term covers two rather different goals. Within personalisation research, these two are commonly distinguished. *Personalised alignment* concerns adapting response behaviour to user-specific preferences that are comparatively explicit (Hong et al., 2024; Li et al., 2024; Meng et al., 2024; Ouyang et al., 2022; Rafailov et al., 2023; Wu et al., 2023; Zhao et al., 2025; Zhou et al., 2024). Much of this literature builds on preference-learning methods originally developed for general LLM alignment. That line of work began with *reinforcement learning from human feedback* (RLHF), which trained models toward explicit general attributes of being helpful, honest, and harmless (Ouyang et al., 2022). The same preferred/rejected supervision was later used to target finer response attributes, such as preferring concise answers, technical explanations, or a particular tone or style, for example a formal register rather than a casual one. In RLHF, a reward model is trained from pairwise preference comparisons and then used to optimise a policy with reinforcement learning (Ouyang et al., 2022). Subsequent work removes the separate RL stage: *direct preference optimisation* (DPO) reparameterises the optimal reward so that the policy can be trained directly on preferred versus rejected responses (Rafailov et al., 2023); *odds-ratio preference optimisation* (ORPO) combines supervised fine-tuning with an odds-ratio preference term and does not require a separate reference model (Hong et al., 2024); and *simple preference optimisation* (SimPO) further simplifies the objective by using a reference-free, length-normalised reward (Meng et al., 2024). When applied to personalisation, these methods steer generation toward an individual user’s stated or elicited preferences rather than toward a single population-level notion of helpfulness (Jang et al., 2023; Kirk et al., 2024; Li et al., 2024; Wu et al., 2023; Zhou et al., 2024). Across this literature, models are therefore aligned *explicitly* to target attributes that are specified through preference labels or instructions, for example

a label marking the shorter of two answers as preferred, or an instruction asking for a beginner-friendly explanation.

However, the attributes needed for personalisation are often not available as explicit labels in the current request. Hence *implicit personalised reasoning*: the relevant preference or constraint must be recovered from earlier interactions. The user typically asks an ordinary question; the personalising signal is latent in prior conversation turns, routines, constraints, or previously revealed facts. For example, a user who asks for a relaxing morning routine may already practise yoga every morning; a personalised assistant should build on that habit rather than returning generic suggestions (Figure 1 develops this example in full). A second example is dietary advice: if earlier dialogue mentioned a nut allergy, a recipe recommendation should respect that constraint even when the current query does not restate it. Doing so requires several steps that go beyond surface style adaptation. The model must locate the relevant past evidence, decide what that evidence implies for the present query, and generate an answer that uses the implication correctly, without fabricating unsupported personal details. This is harder than personalised alignment with explicit preference labels: the supervision is indirect, the evidence may be sparse or noisy, and a plausible generic answer may still fail to satisfy the user’s latent constraint (Jiang et al., 2025; Liu et al., 2025; Zhao et al., 2025). Benchmarks built around long user histories show that even strong LLMs often miss such cues when they are buried among many unrelated turns (Maharana et al., 2024; Wu et al., 2024). This thesis focuses on that setting: given contextual evidence from previous interactions, can a model identify the implicit user information and apply it when answering a new query?

1.4 Retrieval, Memory, and Learning Signals

Building a system for that setting involves two separate decisions: how the relevant evidence is obtained, and how the model is trained to use it once it has been supplied. Existing work on the first is surveyed below, followed by the learning signals available for the second. Systems for this problem typically follow a retrieve-then-generate pattern (Lewis et al., 2020; Salemi et al., 2024b). Given a new query, the system first selects relevant evidence from the user’s history, or from a constructed memory or profile, and then conditions a language model on that evidence to produce a personalised response. Figure 1 shows an example of this pipeline. The user’s history contains many turns; among them is an earlier remark that they practise yoga every morning. Retrieval extracts a conversation snippet that includes this cue (highlighted in the figure). The current query asks for a relaxing morning routine but does not restate the yoga habit. A personalised response should therefore build on that habit, for example by suggesting a short morning yoga sequence, rather than returning a generic routine that ignores what is already known about the user. Existing work addresses different stages of this pipeline.

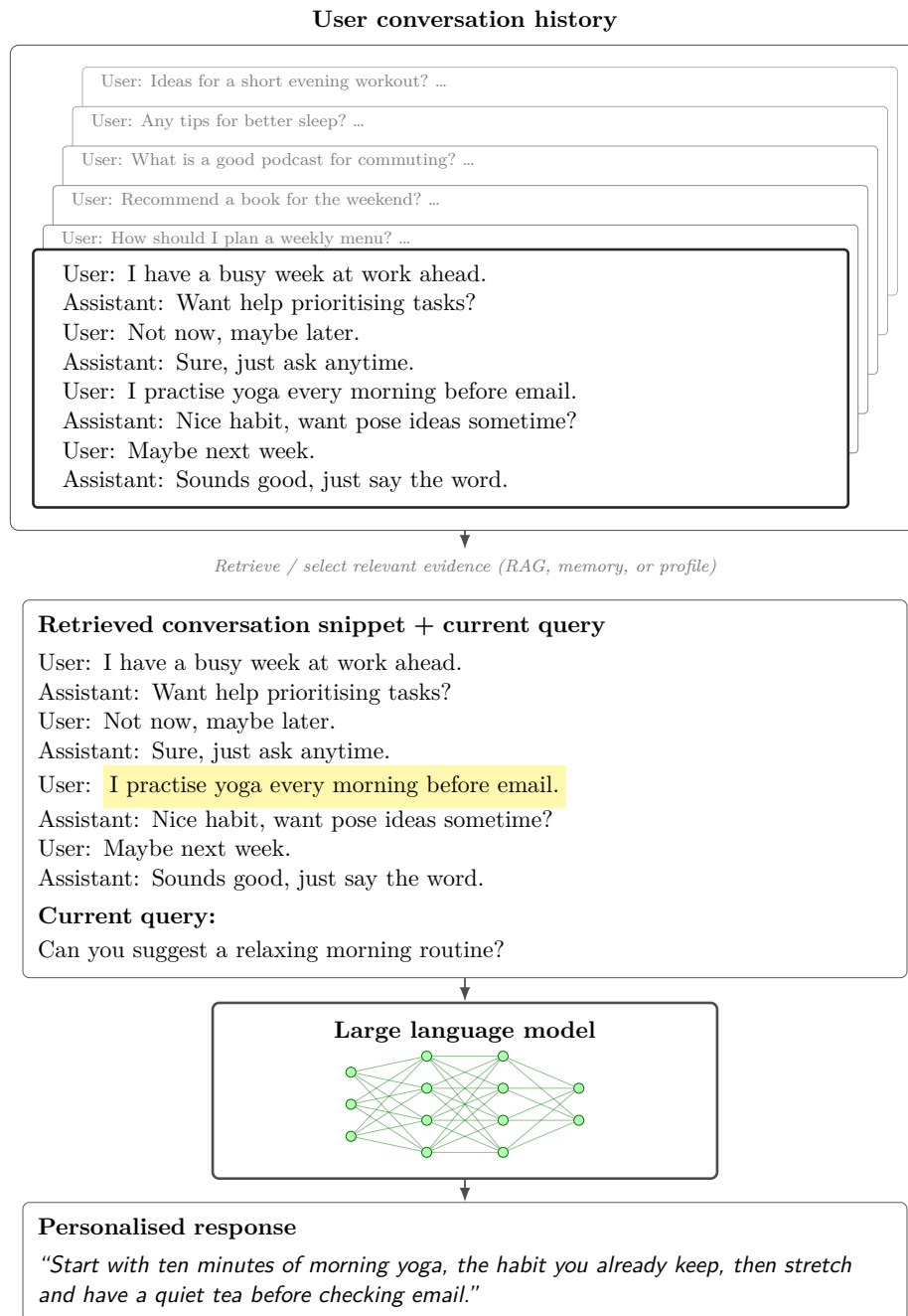


Figure 1: A retrieve-then-generate personalisation pipeline with a concrete example. User history is shown as stacked session cards inside an outer frame: earlier sessions are abbreviated, while the front card shows the most recent session in full. From this history the system selects a snippet that contains the relevant cue (highlighted): the user practises yoga every morning. Given the query for a relaxing morning routine, the language model must use that cue to produce a personalised answer rather than a generic one. This thesis focuses on the model stage, holding retrieval fixed.

Retrieval-augmented methods select useful historical records (Lewis et al., 2020; Liu et al., 2023; Mysore et al., 2024; Richardson et al., 2023; Salemi et al., 2024a,b; Zhuang et al., 2024). Memory systems store, consolidate, and recall user information over time (Chhikara et al., 2025; Dalvi et al., 2022; Jiang et al., 2025; Li et al., 2025; Madaan et al., 2022; Qian et al., 2024; Xu et al., 2026; Zhong et al., 2024; Zhou et al., 2026). These components are complementary: retrieval provides access to specific past interactions, while memory compresses long-term user state.

Given a retrieval or memory component, a second question is how the model is *trained* to use the evidence it receives. Several kinds of learning signal are relevant here. The first ranks responses: preference optimisation raises the likelihood of a preferred answer over a rejected one under a fixed input (Hong et al., 2024; Rafailov et al., 2023). The second abandons ranking and treats a poor answer as material to repair: Chain of Hindsight conditions a model on rated outputs and fine-tunes it to produce the better one (Liu et al., 2024); learning-from-mistakes and self-correction methods fine-tune on mistake-correction pairs in reasoning tasks (An et al., 2023; Qu et al., 2024); and, in personalisation, a separate rewriting module can be trained to convert a generic response into a user-aligned one (Hao et al., 2025). The third uses reinforcement learning to make the model reason about the user before it answers. One line of work learns, by RL, to write a text summary of each user’s preferences, characteristics, and past conversations; the summary conditions the reward model so that it can predict what an individual user values, and summariser and reward model are trained together in an online loop (Nam et al., 2026). Another optimises the answer itself against a personalised reward, training the model to make the implicit intent behind a query explicit before generating a response (Amirizani et al., 2026). The same idea has also been placed inside a retrieval pipeline, where a contrastive reward trains the model to reason explicitly over a *retrieved* user profile in order to recognise the user’s intention (Zhang et al., 2025b). These signals were developed largely for general alignment and reasoning correctness, and where they do target personalisation they rely on reinforcement learning against a learned reward model.

1.5 Scope and Research Question

Retrieving or storing the right evidence does not, by itself, guarantee successful personalisation. Even when the yoga remark is available in the snippet, the model must still recognise that it is the relevant cue for a morning-routine query and generate an answer that uses it, for example recommending yoga, without inventing unsupported personal details. Benchmarks such as PersonaMem-v2 show that strong language models can still struggle to identify and apply such implicit preferences when evidence is buried in long, noisy histories (Jiang et al., 2025; Liu et al., 2025). This thesis therefore concentrates on the second stage in Figure 1: *context-grounded personalised question answering*. Given

a query and contextual evidence drawn from previous interactions, the goal is to learn a model that identifies the relevant implicit user information and generates an appropriately personalised response. We deliberately hold the retrieval interface fixed in order to isolate this reasoning-and-generation problem.

Returning to the privacy motivation above, the same learning problem becomes more demanding under federation. Most studies of implicit personalised reasoning assume centralised access to user histories or personalised training examples during optimisation. By contrast, federated LLM research has so far focused mainly on decentralising alignment and preference optimisation (Fan et al., 2024; Zhu et al., 2026). Considerably less is known about whether a model can acquire a general capability for context-grounded personalised reasoning when the interaction data needed to learn that capability remain distributed across clients. In that regime, each client holds user-specific conversations locally, the server never inspects raw histories, and the global model must improve through aggregated local updates alone.

The research question of this thesis can therefore be stated as follows. Given a user query and a relevant conversation snippet drawn from that user’s history, how can a language model be trained to generate an implicitly personalised answer that correctly uses the evidence in the snippet; and does such learning remain effective when the underlying personalisation data cannot be pooled centrally and must instead be optimised under a federated protocol? The second half of this question is the one least addressed by existing work: objectives of this kind are almost always developed and evaluated under centralised optimisation, so it is unknown whether their relative merits survive when updates are averaged across clients. In other words, we study *implicit personalisation from relevant conversational evidence* under both standard centralised training and a privacy-motivated federated setting, holding retrieval fixed so that the focus remains on reasoning over the provided snippet and generating the answer.

1.6 Approach and Contributions

To answer that question we compare five training objectives experimentally. All five are trained on the same material and differ only in what the model is shown at input and what the loss is computed over. A single training example therefore supplies four things: the user’s query q ; the relevant snippet c^+ drawn from that user’s history; the correctly personalised answer y^+ ; and a *general* answer y^- , a fluent and reasonable reply to q that any user might receive, because it does not draw on the evidence in c^+ . Two of the objectives additionally use a *negative* context c^- , another genuine snippet from the same user’s history that happens not to bear on the current query. This preferred/general pairing follows the framing used in preference learning for personalised alignment (Hong et al., 2024; Li et al., 2024; Meng et al., 2024; Rafailov et al., 2023), applied here to evidence that is implicit rather than stated.

Each objective is best read as a hypothesis about what supervision this capability actually requires, and each is motivated by something the previous one leaves untaught. That is the thread through what follows. The objectives are described informally here and defined precisely in Section 3.

Supervised fine-tuning serves as the baseline. The model is given the query and the relevant snippet, and is trained to produce the personalised answer, which we write $(q, c^+) \rightarrow y^+$: everything to the left of the arrow is what the model reads, and the term to the right is what it is trained to generate. It raises the probability of the correct answer, but it never shows the model what an unpersonalised reply to the same query looks like, so it cannot teach how y^+ differs from y^- , nor which part of c^+ the personalised content rests on.

The *answer-contrastive* objective supplies the first of these. It contrasts y^+ with y^- under the same snippet c^+ , on the hypothesis that preferring the personalised answer over the general one forces the model to notice the contextual cues that separate them. That comparison holds the evidence fixed and varies the answer, so it never tests whether the model is relying on the evidence at all.

The *context-contrastive* objective inverts the comparison: it holds the answer fixed at y^+ and varies the evidence instead, contrasting the relevant snippet c^+ against the uninformative c^- , on the hypothesis that this teaches the model which previous interaction supports the answer. Its blind spot mirrors that of the answer-contrastive objective: it never distinguishes a good answer from a plausible but generic alternative.

The *dual-contrastive* objective combines the two in a single loss, so that answer preference and contextual support act together rather than in isolation.

All four of these rank: they raise the likelihood of one option relative to another. Ranking establishes that y^+ is preferable to y^- , but not how y^- falls short, nor which of its parts should be kept, corrected, or replaced. The *draft-revision* objective abandons ranking and moves y^- from the loss to the input, handing it to the model as a generic draft to be rewritten into y^+ , that is $(q, c^+, y^-) \rightarrow y^+$. Because y^- is a competent answer rather than a broken one, the only work left is to inject what c^+ says about the user. The idea is adapted from Chain of Hindsight (Liu et al., 2024), which conditions a model on a worse output and fine-tunes it to produce a better one; here the revision must be driven by the user-specific cues in c^+ .

All five are trained twice, once centrally and once under a federated protocol, so that the cost of decentralising the personalisation data can be read off directly.

Taken together, the main contributions of this thesis are therefore: (i) a controlled comparison of five training objectives for context-grounded personalised question answering, holding the base model, the data, the retrieval interface, and the training budget fixed, so that differences are attributable to the learning signal alone; (ii) a context-contrastive objective that varies the conditioning evidence instead of the response, applied to non-adversarial negative contexts that are valid user interactions rather than

corrupted or counterfactual ones; (iii) a dual-contrastive objective that combines answer and context contrast in a single loss; and (iv) a draft-revision objective that adapts the learning-from-mistakes concept to evidence-grounded correction, training the model to personalise a general answer against the user’s retrieved history while remaining a single-pass generator at inference. To the best of our knowledge, the context-contrastive, dual-contrastive, and draft-revision objectives have not previously been studied for implicit personalisation, and no prior work evaluates objectives of this kind under both centralised and federated training.

1.7 Thesis Outline

The remainder of this thesis is organised as follows. Section 2 reviews language modelling with Transformer decoders, training and pretraining, inference, instruction fine-tuning and alignment, LoRA, and federated learning of LoRA adapters. Section 3 formalises context-grounded personalised question answering and presents the proposed training objectives in centralised and federated settings. Section 4 reports the experimental findings and analyses. The thesis concludes by summarising the main results, discussing limitations, and outlining directions for future work.

2 Background

This section reviews the concepts needed for the rest of the thesis. We begin with language modelling: how text is tokenised, how an autoregressive model predicts the next token, and how a Transformer decoder implements that prediction. We then explain training, followed by inference. Next we cover the modern LLM pipeline, pretraining, instruction fine-tuning, and preference alignment, then introduce LoRA as a parameter-efficient adaptation method, and close with federated learning, which builds directly on LoRA. Later sections specialise these ideas to personalisation (Section 3) and report experiments (Section 4).

2.1 Language Modelling

A language model assigns probabilities to text and, at generation time, produces new text one unit at a time. Contemporary systems implement this with neural networks that operate on discrete *tokens* rather than on raw characters or words.

Tokenisation. Raw text is first converted into a sequence of tokens by a fixed *tokeniser*. A widely used method is byte-pair encoding (BPE) (Sennrich et al., 2016). BPE proceeds in two stages. First, the training corpus is *broken* into a base alphabet: typically individual characters or bytes, so that every string can be represented exactly. Second,

the algorithm repeatedly *merges* the most frequent adjacent pair of symbols into a new vocabulary entry. For example, if **t** and **h** co-occur often, they form the new token **th**; later merges may produce **the**, **ing**, and other common subwords. Merging stops when a target vocabulary size is reached. The resulting vocabulary $\mathcal{V}$ therefore contains both rare single characters and frequent multi-character units. At encode time, a new string is segmented using the learned merge rules, and each token is mapped to an integer index in $\{1, \dots, |\mathcal{V}|\}$.

This step is essential: neural architectures do not consume strings directly, but embeddings of these discrete IDs. Tokenisation therefore determines both the input representation and the output space over which next-token probabilities are defined. In this thesis, prompts, conversation snippets, and answers are all processed through the same tokeniser before being passed to the model (Section 3).

Autoregressive generation. Let $w = (w_1, \dots, w_n)$ be a token sequence. An *autoregressive* language model factorises its joint probability as

$$p_\theta(w) = \prod_{t=1}^n p_\theta(w_t \mid w_{<t}), \quad w_{<t} = (w_1, \dots, w_{t-1}), \quad (1)$$

where θ denotes the model parameters (Brown et al., 2020; Radford et al., 2019). At each step t , the model predicts a distribution over the vocabulary $\mathcal{V}$ given only the preceding tokens $w_{<t}$. Training and scoring in later sections use the same factorisation: the probability of an answer is the product of these next-token probabilities under a prompt that includes the query and conversation snippet (Section 3).

Transformer decoder. Equation (1) states *what* must be computed, a next-token distribution at each position, but not *how*. In practice, those next-token distributions are produced by a decoder-only *Transformer* (Radford et al., 2019; Vaswani et al., 2017): a neural network that maps the token sequence $w_{<t}$ to a vector of vocabulary logits, which are then turned into probabilities by a softmax. Its central component is the *attention mechanism*, in which each position forms its representation as a weighted combination of the representations at all earlier positions, with the weights computed from those representations themselves. We summarise the path from tokens to those probabilities.

Each token ID is looked up in a learned embedding matrix $E \in \mathbb{R}^{|\mathcal{V}| \times d}$, giving a continuous vector $e_t = E_{w_t} \in \mathbb{R}^d$. Because the attention mechanism alone does not encode order, a positional vector $p_t \in \mathbb{R}^d$ (fixed or learned) is added, giving $h_t^{(0)} = e_t + p_t$. Collecting these vectors as rows yields a matrix $H^{(0)} \in \mathbb{R}^{n \times d}$. These are the representations that subsequent layers mix.

The attention mechanism enables each position to condition on information from other positions in the sequence, which is essential for contextualised next-token pre-

diction. Rather than computing pairwise interactions directly in the embedding space, the Transformer constructs three learned linear projections of the same hidden states $H \in \mathbb{R}^{n \times d}$:

$$Q = HW_Q, \quad K = HW_K, \quad V = HW_V.$$

These projections serve complementary roles in the attention computation. The query matrix Q encodes, for each position, a representation against which other positions are scored for relevance. The key matrix K provides the representations with which those queries are compared via a similarity measure. The value matrix V contains the content that is subsequently aggregated according to the resulting attention weights. Separating relevance scoring (Q, K) from content aggregation (V) allows the model to select which positions to attend to independently of what information those positions contribute. Figure 2 summarises this pipeline from tokens through embeddings to Q, K , and V .

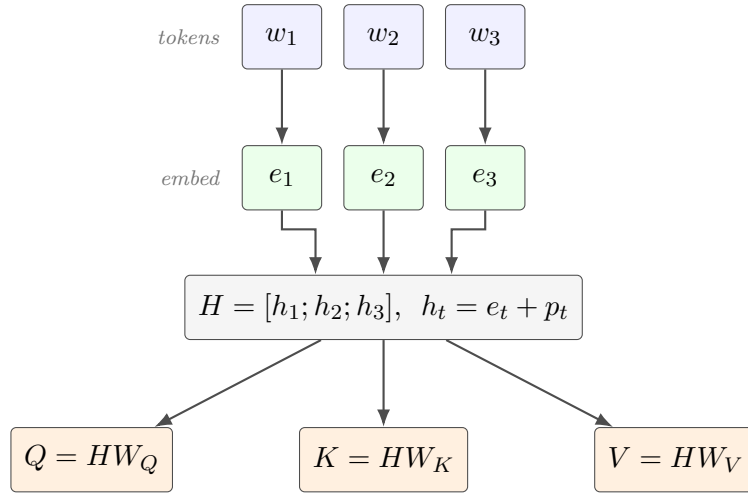


Figure 2: From tokens to attention inputs. Each token w_t is mapped to an embedding e_t , combined with a positional vector to form hidden states H , and then projected into queries Q , keys K , and values V for the attention mechanism.

Scaled dot-product attention mixes information across positions:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

In a causal decoder, future positions are masked so that token t can depend only on $w_{\leq t}$, which preserves the autoregressive factorisation in (1). Multi-head attention repeats this operation with several projections in parallel. Stacked blocks with feed-forward layers, residual connections, and normalisation produce contextual representations; a final linear head and softmax yield the next-token distribution $p_\theta(\cdot \mid w_{< t})$.

2.2 Training

Training adjusts θ so that the model assigns high probability to desired token sequences. Each training example is a token sequence of the kind described in (1), split into two parts: writing $w = (x, z)$ for the full sequence, x is the conditioning prompt, possibly empty, and z is the target completion the model is trained to predict. In practice training minimises an average loss over a dataset of N such prompt–completion pairs,

$$\hat{\theta} \in \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell(\theta; x_i, z_i).$$

For autoregressive models the standard loss is token-level cross-entropy, i.e. the negative log-likelihood of the target tokens,

$$\ell(\theta; x, z) = - \sum_{t=1}^{|z|} \log p_{\theta}(z_t \mid x, z_{<t}). \quad (2)$$

The distribution p_{θ} appearing here is the same next-token distribution as in (1); only the notation for its argument has changed. Conditioning on $(x, z_{<t})$ conditions on exactly the tokens preceding z_t in w , so (2) is the negative logarithm of those factors of (1) that correspond to completion positions. The prompt is written separately from the completion because the sum runs over z alone: in practice prompt tokens are masked so that they contribute no gradient.

Optimisation in practice. Closed-form solutions are unavailable for models of this size. Training therefore uses mini-batch stochastic gradient methods: at each step a batch of examples is sampled, the loss and its gradient are computed, and parameters are updated with an optimiser such as Adam or AdamW (Loshchilov et al., 2017). Important practical choices include the learning rate and its schedule, the batch size, the number of epochs (passes over the data), weight decay, and gradient clipping. These hyperparameters strongly affect stability and final quality; in our experiments they are matched across methods where possible so that comparisons reflect the training objective rather than tuning artefacts (Section 4).

2.3 Inference

Once parameters are fixed, *inference* produces text by autoregressive decoding. Starting from a prompt (for example a system message, conversation snippet, and user query), the model computes $p_{\theta}(\cdot \mid w_{<t})$ and selects the next token by greedy argmax or by sampling from the distribution (often with temperature or top- p truncation). The chosen token is appended to the context and the process repeats until an end-of-sequence token or a

length limit is reached. The resulting string is the model output.

In this thesis, free-form personalised answers are generated in this way, while the multiple-choice metric instead scores candidate answers by length-normalised likelihood under the same prompt without requiring open-ended decoding (Section 3).

2.4 Large Language Models

A *large language model* is a Transformer decoder trained at scale and then adapted into an assistant. We summarise the stages that produce one and that are relevant here.

Pretraining. The first large-scale application of the training objective of Section 2.2 is *pretraining*. A Transformer decoder is trained with the next-token loss (2) on a very large text corpus (web pages, books, code, and related sources) (Brown et al., 2020; Radford et al., 2019). At each step, a segment of tokens is passed through the Transformer; the model predicts the distribution over the next token at every position; the cross-entropy between that prediction and the true next token is averaged; and gradients of this loss are used to update θ . The goal is general language competence, fluency, world knowledge, and basic reasoning patterns, rather than a single downstream task. Scaling this procedure yields models with broad capabilities (Brown et al., 2020; Radford et al., 2019). We treat the resulting pretrained base model as given and do not re-run this stage in our experiments.

Instruction fine-tuning. Pretraining optimises next-token prediction on unstructured text, so a pretrained model tends to continue a prompt rather than respond to it: presented with a question, it may carry on with further questions in the same style instead of answering. Turning such a model into a useful assistant therefore requires further stages. *Supervised fine-tuning* (SFT) adapts a pretrained model on curated prompt–response pairs so that it answers questions and follows task formats (Ouyang et al., 2022; Wei et al., 2022). The loss remains next-token cross-entropy on the response tokens, with the prompt masked, as in (2). In practice, the fine-tuning corpus mixes many task types rather than a single format: question answering, summarisation, translation, classification or information extraction, and open-ended dialogue or instruction following, among others. Training on this mixture teaches the model to interpret a wide range of prompts, not only to continue unrestricted text. Instruction-tuned models such as those in the GPT (Ouyang et al., 2022), Llama (Touvron et al., 2023), and Qwen (Yang et al., 2025) families are widely used as general assistants. Our experiments start from a small Qwen instruction-capable base (Yang et al., 2025) and further adapt it for personalised answering (Section 4).

2.5 Low-Rank Adaptation (LoRA)

Full fine-tuning updates every parameter of the Transformer and is expensive in memory and, when training is distributed, in communication. After instruction fine-tuning (or starting from an instruction-tuned base), downstream adaptation is therefore often performed with *low-rank adaptation* (LoRA) (Hu et al., 2021). LoRA freezes the base weights W_0 and trains only a low-rank update

$$W = W_0 + BA, \quad B \in \mathbb{R}^{d \times r}, \quad A \in \mathbb{R}^{r \times k}, \quad r \ll \min(d, k).$$

Only A and B (and any other adapter parameters) are optimised. The resulting adapter is small relative to the base model, which makes repeated fine-tuning practical and, as discussed next, makes federated updates cheap to communicate. In our setting, both supervised and preference-based personalisation objectives are applied to LoRA adapters rather than to the full base model. All centralised and federated runs in this thesis fine-tune LoRA adapters on top of a frozen base model (Sections 3 and 4).

2.6 Federated Learning

Federated learning trains one shared model on data that never leaves the devices holding it (Kairouz and McMahan, 2021; McMahan et al., 2017). A federation has K *clients* and one coordinating *server*. Client k holds a private dataset $\mathcal{D}_k$ of n_k examples, and the federation minimises

$$F(\theta) = \sum_{k=1}^K p_k F_k(\theta), \quad F_k(\theta) = \frac{1}{n_k} \sum_{\xi \in \mathcal{D}_k} \ell(\theta; \xi), \quad p_k = \frac{n_k}{n},$$

where F_k is the loss on client k alone, $n = \sum_k n_k$, and p_k weights each client by how much data it holds. Pooling the data and training centrally would minimise the same F . The difference is a constraint, not a different goal: no client may see another client's data, and the server sees none at all.

Training runs in synchronous *rounds*, and each round has three steps. The server sends the current parameters $\theta^{(t)}$ to every client. Each client then trains on its own data and returns an update $\Delta_k^{(t)}$, which may be either the gradient it computed or the weights it arrived at after training locally. Finally the server combines these updates, usually by the weighted average $\sum_k p_k \Delta_k^{(t)}$, to form $\theta^{(t+1)}$, and the next round begins. Methods differ mainly in how much work a client does before it reports back. At one extreme the client computes the gradient on a single mini-batch and sends it immediately, without changing its own copy of the model; this is FedSGD. At the other it makes several passes over its whole dataset, updating its copy as it goes, and sends the weights it ends up with; this is FedAvg (McMahan et al., 2017). More local work means fewer rounds, and

so less communication, but the client copies drift further apart before they are averaged.

For deep networks the cost of this protocol is dominated by communication rather than computation. Every round moves a quantity the size of θ in each direction, out to each client and back again, so the parameter count is the per-round bandwidth bill. For a model with billions of parameters that runs to gigabytes per client per round, which is untenable over ordinary connections.

LoRA (Section 2.5) removes the barrier. The frozen base weights W_0 are identical on every client and never change during training, so they need never be sent; only the small adapter has to be federated (Jiaxing et al., 2024; Tan et al., 2024; Zhang et al., 2024a). What travels each round is then the adapter alone, orders of magnitude smaller than the full model. We therefore federate the adapter, and we do so with FedGradAvg, the variant in which clients return gradients rather than locally trained weights. This is the FedSGD baseline introduced alongside FedAvg (McMahan et al., 2017), and more broadly synchronous distributed SGD (Chen et al., 2016). Section 3 specifies it in full and contrasts federated with centralised training under the same objectives.

In summary, this section has introduced tokenised autoregressive modelling with Transformer decoders, pretraining and inference, instruction fine-tuning and alignment, LoRA, and federated optimisation of LoRA adapters. Section 3 specialises these tools to context-grounded personalised question answering.

3 Methods

This section presents the methodology developed to answer the research question of this thesis. Before the individual methods are introduced, the problem itself is restated, so that the design choices that follow can be read against it.

The setting studied in this thesis is the following. A user’s conversations with an assistant accumulate over time into a personal history. When the user asks a new question, a retriever selects a snippet of that history and places it in the prompt alongside the question, and the model is expected to produce an answer consistent with what the snippet indicates about the user. This is the task of *context-grounded personalised question answering*: the information needed to answer well is never stated in the question itself, so the model has to notice which part of the supplied conversation bears on the current request, work out what it implies for the answer, and use that implication without inventing personal details the evidence does not support. The retrieval stage is held fixed throughout this work, so the model is always given a context that contains the relevant snippet, and the question under study is whether the model can exploit it. The histories that make personalisation possible are also the most sensitive data a user holds, since they record daily routines, relationships, health, and work. Collecting them into a single training corpus is therefore undesirable, and in regulated or on-device

deployments it is not permitted at all. Each objective is consequently studied under two training regimes: a centralised one, in which all user data are pooled on a server, and a federated one, in which the raw histories never leave the user’s device and only model updates are exchanged with the server.

Recognising what a snippet indicates about a user, and acting on it, is a capability that must be trained. Section 2.2 described training as the minimisation of a loss over the training data; once the base model and the data are fixed, the objective determines which behaviour the model acquires. The choice of objective is therefore the design variable examined in this section.

Preference learning provides a natural starting point, since alignment already supplies machinery for training a model to favour one answer over another (Section 2.4). That machinery was developed for *explicit* alignment, where the property an answer should satisfy is stated in the supervision signal. In the present setting the property is implicit: it must first be recovered from the context before it can be satisfied, and it is not clear which of these learning signals transfer under that change. We therefore explore what a training objective should place side by side, and what the model should learn from the difference: two answers, one preferred and one not; two contexts, one that supports the preferred answer and one that does not; both pairs at once; or a signal that abandons comparison altogether. Each objective is a hypothesis about the supervision that this capability requires, and each is evaluated under both centralised and federated training so that the effect of decentralising the data can be separated from the effect of the objective itself.

Three of these objectives are methodological contributions of this section, and each extends existing alignment machinery to implicit personalisation. First, we introduce a context-contrastive signal that varies the conditioning evidence rather than the response, so that supervision acts on *where* the personalised content comes from rather than only on which answer is preferred. Second, we combine answer and context contrast in a single dual-contrastive objective, which supervises the preferred answer and its supporting evidence jointly. Third, we depart from ranking altogether and propose a draft-revision objective: instead of pushing down the likelihood of a general answer, the model receives it as a draft and is trained to personalise it against the user evidence, turning the non-personalised response from a negative example into corrective supervision.

Building on the autoregressive Transformer, SFT, LoRA, and federated LoRA background in Section 2, the remainder of the section is organised as follows. We first introduce the problem setting and mathematical notation, then present the approaches in sequence, beginning with supervised fine-tuning (SFT) as a baseline and proceeding through the answer-contrastive, context-contrastive, dual-contrastive, and draft-revision objectives. We then describe the use of these objectives in centralised and federated settings, and finally present the core assessment metrics: LLM-as-a-judge personalisation success rate and multiple-choice answer accuracy. Exact prompt templates are deferred

to Appendix 1.

3.1 Problem Setting and Notation

We consider the problem of generating a response that answers a user query while remaining consistent with user-specific contextual evidence drawn from previous interactions.

Let $\mathcal{Q}$ denote the space of queries, $\mathcal{C}$ the space of contexts, and $\mathcal{Y}$ the space of textual responses. A query–context pair is written

$$x = (q, c) \in \mathcal{Q} \times \mathcal{C},$$

where $q \in \mathcal{Q}$ is the current user query and $c \in \mathcal{C}$ is contextual information associated with the user (e.g. a previous interaction that may contain a relevant preference, habit, need, or constraint).

We distinguish two types of context:

- c^+ : a *positive* (relevant) context that contains the user-specific information required to produce the preferred response;
- c^- : a *negative* context that is a valid interaction from the user’s history but does not contain information relevant to producing the preferred response.

Datasets of this kind do not always supply both contexts. The dataset used in this work provides a positive context c^+ for each example but no negative one, so c^- is not given and must be constructed from the user’s own history. The procedure used to construct it is described in Section 3.2.4, where the objective that requires it is introduced. Likewise, for responses we distinguish:

- y^+ : the *preferred* (correctly personalised) response;
- y^- : a *general* (non-personalised) response, i.e. a plausible answer to q that could be given to any user because it does not reflect the user-specific evidence in c^+ .

This preferred/general pairing is inspired by the preference-learning formulation used in personalised alignment, where models are trained to favour a preferred completion over a rejected alternative (Hong et al., 2024; Li et al., 2024; Meng et al., 2024; Rafailov et al., 2023). Here the same idea is applied to *implicit* personalisation: y^+ is consistent with the user-specific evidence in c^+ , while y^- answers the same query without using that evidence. Note that y^- is not an erroneous or off-topic response; it is what a competent but unpersonalised assistant would say. The two objectives that consume y^- exploit this differently: the answer-contrastive objective treats it as the rejected side of a preference pair, whereas the draft-revision objective treats it as a generic draft to be

personalised. The preferred response y^+ is expected to answer q while being consistent with the information in c^+ .

Let

$$\mathcal{D} = \{(q_i, c_i^+, c_i^-, y_i^+, y_i^-)\}_{i=1}^N \quad (3)$$

denote the set of training examples. Not every objective uses all five fields: SFT uses only (q, c^+, y^+) , the answer-contrastive objective uses (q, c^+, y^+, y^-) , the context-contrastive objective uses (q, c^+, c^-, y^+) , and the draft-revision objective uses (q, c^+, y^-) as input with target y^+ .

Model. The goal is to learn a conditional language model

$$\pi_\theta : \mathcal{Q} \times \mathcal{C} \rightarrow \mathcal{P}(\mathcal{Y}),$$

where $\mathcal{P}(\mathcal{Y})$ denotes the set of probability distributions over $\mathcal{Y}$ and θ denotes the trainable parameters. Given (q, c) , the model assigns probability $\pi_\theta(y \mid q, c)$ to each candidate response $y \in \mathcal{Y}$.

Learning objective. Parameters are estimated by minimising an empirical risk

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(\theta; q_i, c_i^+, c_i^-, y_i^+, y_i^-),$$

where the loss $\mathcal{L}$ depends on the chosen training method. At inference, the model receives (q, c^+) and generates a response $\hat{y}$ by autoregressive decoding (Section 2.3).

3.2 Approaches

As described in Section 2.4, fine-tuning is how a pretrained language model is directed towards a particular behaviour: the training objective decides which responses are reinforced and which are discouraged, and so determines what the model does with the input it receives. The behaviour required here is personalised answering, in which the model must recover the facts about the user that are contained in the retrieved snippet and use them to answer the query instead of replying generically. As the objective is what shapes that behaviour, it is the variable this section studies.

Five objectives are compared: supervised fine-tuning (SFT), which serves as the trained baseline, followed by the answer-contrastive, context-contrastive, dual-contrastive, and draft-revision objectives. They differ in which fields of a training example (3) the loss consumes. Preceding all of them is a zero-shot reference point that involves no training at all and simply prompts the pretrained model, so that the effect of each objective can be read against what the model can already do.

Everything else is held fixed so that any difference in behaviour is attributable to the learning signal alone. All five objectives fine-tune the same base model, Qwen3-0.6B (Yang et al., 2025), for three epochs with a learning rate of 2×10^{-4} . The prompting structure is shared throughout, namely a system instruction to use the provided context followed by a user message containing the context and the query (Appendix 1), and supervision is teacher-forced on the target response tokens in every case. The model is small on purpose. At 0.6B parameters it can run on a personal device, which is the setting this thesis targets: if a user’s history is not to leave the device, the model that reads it should live there too. It is also practical, since the ten runs (five objectives in two regimes) are trained one at a time, and each run has to fit the model, its optimiser state, and its training batches on a single Google Colab GPU and finish within one session.

The zero-shot reference is described first, after which the objectives are introduced in the order in which they were developed, each one motivated by a limitation of the one before it.

3.2.1 Zero-Shot Prompting

The simplest way to answer a personalised query is not to train at all. In the zero-shot setting the pretrained model is given the same prompt as every other method, namely the system instruction, the positive context c^+ , and the query q (Appendix 1), and an answer is decoded from it directly, using the pretrained parameters θ_0 in place of the fine-tuned θ^* . No parameters are updated and no personalised example is ever shown.

This is a meaningful starting point because the base model is not naive. Pretraining followed by instruction tuning on a broad mixture of tasks (Section 2.4) leaves the base model fluent, able to follow instructions, and able to read a conversation snippet placed in its prompt, so it can already produce a plausible answer to q . What is unknown is how far that generic competence goes when the answer must also reflect the user-specific evidence in c^+ . The zero-shot condition isolates exactly this: it measures what supplying the context in the prompt buys on its own, with no training signal about how the context should be used, and any improvement reported later is therefore attributable to the objective rather than to the retrieval interface.

Its weakness is the reason the remaining approaches exist. A model that has never seen a personalised target has no way of learning that the implicit preference in c^+ , and not generic advice, is what the answer is supposed to reflect. Supervised fine-tuning is the natural stronger baseline, since it teaches that behaviour in the most direct way available, by imitation of the preferred answer.

3.2.2 Supervised Fine-Tuning (SFT)

The first trained approach is supervised fine-tuning. As described in Section 2.4, instruction fine-tuning is the standard way to adapt a pretrained model to a particular task: the model is trained on prompt–response pairs using the next-token cross-entropy loss on the response, with the prompt masked. Here the task is context-grounded personalised answering, so the prompt carries the query q together with the positive context c^+ , and the target is the preferred answer y^+ .

For a response $y = (y_1, \dots, y_{|y|})$ under a prompt (q, c) , write the *length-normalised* log-likelihood as

$$L_\theta(y \mid q, c) = \frac{1}{|y|} \sum_{t=1}^{|y|} \log \pi_\theta(y_t \mid q, c, y_{<t}), \quad y_{<t} = (y_1, \dots, y_{t-1}). \quad (4)$$

Only response tokens contribute to (4), and dividing by the response length keeps the quantity comparable across answers of different lengths, which matters as soon as two candidate responses are compared. This quantity is reused by every objective below.

The SFT loss is the negative of (4) evaluated at the preferred answer under the positive context,

$$\mathcal{L}_{\text{SFT}}(\theta) = -L_\theta(y^+ \mid q, c^+) = -\frac{1}{|y^+|} \sum_{t=1}^{|y^+|} \log \pi_\theta(y_t^+ \mid q, c^+, y_{<t}^+).$$

Prompt tokens are masked, so only answer tokens contribute. Neither y^- nor c^- is used. SFT increases the probability of generating y^+ given q and c^+ , but provides no explicit signal about how y^+ differs from an insufficiently personalised answer, or which parts of the supplied context support the personalised content.

3.2.3 Answer-Contrastive Objective

SFT only ever sees personalised answers, so it is never shown what a general, non-personalised answer to the same query looks like. The training data, however, also contain a general answer y^- for each example, and the second approach asks whether exposing the model to that answer under the same relevant context helps it identify which aspects of c^+ matter for personalisation. The hypothesis is that increasing $\pi_\theta(y^+ \mid q, c^+)$ relative to $\pi_\theta(y^- \mid q, c^+)$ forces the model to recognise the contextual cues that distinguish a correctly personalised response from a generic one. If y^- fails to use the user-specific information in c^+ , while y^+ succeeds, then ranking these answers under a fixed context should highlight the relevant content in c^+ , which is a signal SFT cannot provide.

We implement this comparison with odds-ratio preference optimisation (ORPO) (Hong et al., 2024). ORPO was proposed as a reference-free alternative to RLHF (Ouyang et al.,

2022) and DPO (Rafailov et al., 2023): instead of training a reward model or holding a frozen reference policy in memory, it appends a single odds-ratio penalty to the supervised loss, so that instruction fine-tuning and preference learning occur in one stage. Hong et al. (2024) argue that contrasting the *odds* of the two responses, rather than their probabilities, gives a milder and better conditioned penalty: it separates the rejected answer from the preferred one without driving its likelihood down so aggressively that generation quality degrades. Both properties suit the present setting, where the SFT term must remain in place to teach the model to produce y^+ at all, and the contrast is added on top of it.

For a response y under a prompt (q, c) , the odds under the model are

$$\text{odds}_\theta(y \mid q, c) = \frac{\exp(L_\theta(y \mid q, c))}{1 - \exp(L_\theta(y \mid q, c))},$$

or equivalently, in logarithmic form,

$$\log \text{odds}_\theta(y \mid q, c) = L_\theta(y \mid q, c) - \log(1 - e^{L_\theta(y \mid q, c)}),$$

with L_θ the length-normalised log-likelihood of (4). Let $\sigma(z) = 1/(1 + e^{-z})$ denote the logistic sigmoid.

The answer-contrastive objective therefore uses (q, c^+, y^+, y^-) . Define the answer-contrastive odds-ratio term

$$\mathcal{L}_{\text{ans}}(\theta) = -\log \sigma\left(\log \text{odds}_\theta(y^+ \mid q, c^+) - \log \text{odds}_\theta(y^- \mid q, c^+)\right). \quad (5)$$

The full training objective is

$$\begin{aligned} \mathcal{L}_{\text{AC}}(\theta) &= \mathcal{L}_{\text{SFT}}(\theta) + \lambda_y \mathcal{L}_{\text{ans}}(\theta) \\ &= -L_\theta(y^+ \mid q, c^+) - \lambda_y \log \sigma\left(\log \text{odds}_\theta(y^+ \mid q, c^+) - \log \text{odds}_\theta(y^- \mid q, c^+)\right), \end{aligned}$$

where $\lambda_y > 0$ controls the strength of preference alignment. Minimising $\mathcal{L}_{\text{AC}}$ increases the relative odds of y^+ over y^- while still fitting the preferred completion. This objective teaches the model which response is preferable under c^+ , but still does not vary the context itself, so contextual relevance is only inferred indirectly through answer quality.

3.2.4 Context-Contrastive Objective

To supervise contextual relevance more directly, we introduce a negative context c^- as an explicit source of contrast. The hypothesis is that increasing $\pi_\theta(y^+ \mid q, c^+)$ relative to $\pi_\theta(y^+ \mid q, c^-)$ teaches the model that the personalised answer is supported by the relevant interaction and not by an unrelated one. Holding the response fixed and varying the

context isolates which previous interaction provides the necessary user information. As above, the comparison is implemented through the odds ratio of the length-normalised likelihoods.

The context-contrastive objective therefore uses (q, c^+, c^-, y^+) . The positive context c^+ contains the specific user information required for the current query. The negative context c^- is also a valid interaction from the user’s history, but it does not provide evidence for the personalised content of y^+ . It is negative because it does not support y^+ , rather than because it contains false information.

Analogously to (5), define the context-contrastive term

$$\mathcal{L}_{\text{ctx}}(\theta) = -\log \sigma\left(\log \text{odds}_{\theta}(y^+ | q, c^+) - \log \text{odds}_{\theta}(y^+ | q, c^-)\right). \quad (6)$$

The training objective is then $\mathcal{L}_{\text{CC}}(\theta) = \mathcal{L}_{\text{SFT}}(\theta) + \lambda_c \mathcal{L}_{\text{ctx}}(\theta)$, where $\lambda_c > 0$ and $\mathcal{L}_{\text{SFT}}$ is computed under c^+ . This encourages the model to assign higher odds to y^+ when conditioned on relevant context than when conditioned on non-informative context.

Constructing the negative context. The loss (6) depends only on the pair (c^+, c^-) together with (q, y^+) , but the choice of c^- decides what the contrast actually teaches. We take it from the user’s own history. Let $H = (h_1, \dots, h_M)$ be that history as a sequence of turns, with any system turn discarded so that no instruction text can leak into the negative context. The first step is to find where c^+ sits inside H , by matching its sequence of turns against H . The position is needed only so that the positive evidence can be excluded from the pool of candidates, since a window overlapping it would contain the very information the negative context is supposed to lack. The history is then cut into consecutive windows of $k = \max(2, |c^+|)$ turns, where $|c^+|$ is the number of turns in the positive snippet, and any window overlapping the located span is dropped; matching the window length to $|c^+|$ keeps context length from becoming a confound in the comparison. Each remaining window is scored by the cosine similarity between its TF-IDF (Robertson, 2004) representation and that of c^+ , with a token-level Jaccard overlap as a fallback, and c^- is drawn uniformly, under a fixed seed, from the three least similar windows. When the history is too short to supply a non-overlapping window, the overlap constraint is relaxed and any window of the required length may be selected.

This construction gives c^- three properties that matter for the objective. It is a genuine past interaction of the same user rather than corrupted or synthetic text, so the model is asked to separate relevant evidence from irrelevant evidence and not real text from fabricated text. It is topically distant from c^+ , so it is unlikely to carry the user information on which the preferred answer depends. And it never leaves the user’s own data, so a client can build it without seeing any other client’s history. Drawing from the three most off-topic windows rather than always taking the single most off-topic one introduces variety across examples. The criterion is lexical and therefore only a proxy:

low word overlap makes it unlikely, though not impossible, that a window expresses the same preference as c^+ .

3.2.5 Dual-Contrastive Objective

As we will see in Section 4, neither the answer-contrastive nor the context-contrastive objective alone improves upon supervised fine-tuning. This suggests that each mechanism is incomplete on its own: contrasting answers under a fixed context does not explicitly enforce sensitivity to the source of personal information, while contrasting contexts under a fixed answer does not directly distinguish a preferred response from a plausible alternative.

We therefore combine the two complementary signals using the terms already defined above:

$$\mathcal{L}_{\text{dual}}(\theta) = \mathcal{L}_{\text{SFT}}(\theta) + \lambda_y \mathcal{L}_{\text{ans}}(\theta) + \lambda_c \mathcal{L}_{\text{ctx}}(\theta),$$

with $\lambda_y, \lambda_c > 0$. In our experiments the two preference weights are set equal, $\lambda_y = \lambda_c = \lambda$. Each example uses three length-normalised likelihoods,

$$L_\theta(y^+ | q, c^+), \quad L_\theta(y^+ | q, c^-), \quad L_\theta(y^- | q, c^+),$$

and pushes the model both to prefer y^+ over y^- under c^+ , and to assign higher odds to y^+ under c^+ than under c^- . The dual-contrastive method teaches the model what to generate, which answer is preferable, and which previous interaction provides the evidence for that answer. In contrast to the individual objectives, this joint supervision improves upon the SFT baseline (Section 4), indicating that answer preference and contextual support are complementary.

3.2.6 Draft-Revision Objective

The dual-contrastive result raises a further question: if y^- is informative, is probability ranking the most effective way to use it? In the answer-contrastive and dual-contrastive objectives, y^- appears only as a rejected candidate whose likelihood is driven below that of y^+ . This tells the model that one answer is better than another, but not *how* the rejected answer fails relative to the context, nor which of its parts should be retained, corrected, or replaced.

The draft-revision objective therefore uses y^- differently. Instead of treating the rejected response solely as a ranking target, we present it as an input draft and train the model to revise it into the correct personalised response:

$$(q, c^+, y^-) \longrightarrow y^+.$$

The draft y^- is a general answer rather than a broken one, so this mapping isolates

personalisation itself: it already answers q acceptably, and the only work left for the model is to inject the user-specific evidence in c^+ . Given q , c^+ , and the draft y^- , the model must identify which parts of the draft are generic where the context calls for something user-specific, and rewrite exactly those parts. In this sense, y^- is no longer only a negative example to be down-weighted; it becomes a corrective curriculum that teaches the model how a generic answer differs from a personalised one and how to close that gap using the relevant user evidence. Learning from a worse output is an established idea, but what counts as an improvement differs across settings. Learning-from-mistakes and self-correction methods fine-tune on mistake–correction pairs to improve reasoning, typically in domains such as mathematics, where the correction is judged against a known answer (An et al., 2023; Qu et al., 2024). Chain of Hindsight (Liu et al., 2024) instead addresses explicit personalisation: the model is conditioned on rated outputs and learns to produce the better one, where better means better suited to the person along a stated dimension, such as a simpler or a more technical answer, and no user history is involved. Closest to the present setting is the post-hoc rewriting framework of Hao et al. (2025), which is also implicit, but keeps generation and rewriting as two separate stages at both training and inference time. The draft-revision objective differs on this last point: revision is one of two tasks in a single multi-task training objective, so the corrective signal is absorbed into the parameters of one model and inference remains single-pass.

Concretely, the draft-revision objective is a multi-task SFT objective rather than an odds-ratio preference loss. It trains the model on two input formats that share the same target y^+ :

1. **Generation:** $(q, c^+) \longrightarrow y^+$, identical to SFT, with loss $\mathcal{L}_{\text{gen}}(\theta) = \mathcal{L}_{\text{SFT}}(\theta)$;
2. **Revision:** $(q, c^+, y^-) \longrightarrow y^+$, where y^- is provided as a draft answer.

Both tasks use the length-normalised next-token loss on y^+ ; only the prompt differs. The revision prompt asks the model to revise the draft so that it is consistent with c^+ (Appendix 1). Writing $L_{\theta}^{\text{rev}}(y^+ | q, c^+, y^-)$ for the length-normalised log-likelihood of y^+ under the revision prompt, define

$$\mathcal{L}_{\text{rev}}(\theta) = -L_{\theta}^{\text{rev}}(y^+ | q, c^+, y^-).$$

The draft-revision objective is then

$$\mathcal{L}_{\text{DR}}(\theta) = \mathcal{L}_{\text{gen}}(\theta) + \lambda_{\text{DR}} \mathcal{L}_{\text{rev}}(\theta),$$

where $\lambda_{\text{DR}} > 0$ balances the two tasks. At evaluation, single-pass generation with the generation prompt is used, matching the other methods. As we will see in Section 4,

$\mathcal{L}_{\text{DR}}$ outperforms supervised fine-tuning, the individual answer-contrastive and context-contrastive objectives, and the dual-contrastive objective.

3.3 Centralised and Federated Training

All objectives above are evaluated in both centralised and federated settings. This enables a controlled comparison of how decentralising the personalised training data affects learning of implicit, context-grounded personalisation.

Centralised setting. In the centralised setting, all training examples in $\mathcal{D}$ are pooled on a single server. Optimisation of the objectives defined in Section 3.2 proceeds with standard mini-batch stochastic gradient descent (or an adaptive variant) over the full dataset. This setting provides an upper-reference for what each objective can achieve when user-specific interaction data can be inspected jointly during training.

Federated setting. In the federated setting, user-specific interaction data remain distributed across clients, in the client-server formulation of Section 2.6. Let there be K clients, where client k holds a local dataset $\mathcal{D}_k \subseteq \mathcal{D}$ associated with one or more users (in this work, one user per client), and $\bigcup_{k=1}^K \mathcal{D}_k = \mathcal{D}$ with pairwise disjoint $\mathcal{D}_k$. The central server never receives raw conversational histories; instead, clients compute local gradients on private batches and communicate only those gradients (or equivalently, LoRA parameter updates), which are aggregated to update a shared global model. Figure 3 summarises this protocol.

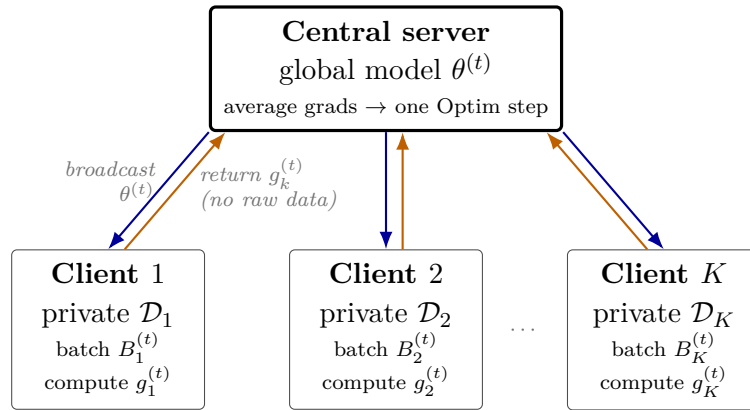


Figure 3: Federated gradient-averaging (FedGradAvg) used in this thesis. At each step the server broadcasts the current global parameters $\theta^{(t)}$. Every client computes a gradient $g_k^{(t)}$ of the chosen personalisation objective on one private mini-batch and returns only that gradient. The server averages the client gradients and applies one global optimiser update. Raw conversation histories never leave the clients.

Training follows a *gradient-averaging* protocol (FedGradAvg / FedSGD) (McMahan et al., 2017), rather than FedAvg-style local optimisation and weight averaging. There is a single global model θ and a single global optimiser whose state persists across the run. Clients perform no local optimiser steps and no local epochs: at each synchronous step they only evaluate the chosen objective $\mathcal{L}$ and return a gradient. Concretely, at global step t :

1. the server broadcasts the current global parameters $\theta^{(t)}$ to all active clients;
2. each client k draws one local mini-batch $B_k^{(t)} \subset \mathcal{D}_k$ and computes the batch gradient

$$g_k^{(t)} = \nabla_{\theta} \frac{1}{|B_k^{(t)}|} \sum_{x \in B_k^{(t)}} \mathcal{L}(\theta^{(t)}; x)$$

on the shared global weights;

3. the server averages the per-client gradients with equal client weight,

$$\bar{g}^{(t)} = \frac{1}{K} \sum_{k=1}^K g_k^{(t)},$$

optionally clips $\bar{g}^{(t)}$, and applies one global parameter update

$$\theta^{(t+1)} = \theta^{(t)} - \eta \bar{g}^{(t)}, \quad (7)$$

where $\eta > 0$ is the learning rate. In practice we use an adaptive optimiser (AdamW) with the same averaged gradient $\bar{g}^{(t)}$ in place of the plain step (7).

To keep communication and on-device computation tractable for language models, trainable updates are restricted to parameter-efficient modules (e.g. LoRA adapters), while the base model remains frozen. The same local objective $\mathcal{L}$ is used as in the centralised setting; only the optimisation topology changes. This design isolates the effect of decentralised, gradient-aggregated optimisation on learning context-grounded personalised reasoning.

3.4 Evaluation Metrics

What we wish to measure is whether a trained model uses the user-specific evidence it is given, that is, whether its answer to a query reflects the preference recorded in the accompanying context rather than the generic answer any model would produce. This is harder to quantify than standard accuracy, because a personalised answer is free-form text: there are many ways to phrase a good answer and surface-overlap scores such as BLEU (Papineni et al., 2002) or ROUGE (Lin, 2004) reward wording rather than

whether the user’s preference was honoured. Two evaluation protocols are established in the personalised and aligned generation literature for exactly this reason: judging generated text with a frontier model (Zheng et al., 2023), and scoring a fixed set of candidate answers by their likelihood under the model (Jiang et al., 2025; Salemi et al., 2024b). We adopt both. They are complementary: the first tests whether the model can *write* a correctly personalised answer, and the second whether it can *recognise* one. Both operate on the same held-out test set of N_{test} examples with known preferred responses y^+ , and practical details, including prompt templates, are given in Appendix 1.

3.4.1 LLM-as-a-Judge Personalisation Success Rate

The first metric evaluates what the deployed system actually produces: a free-form answer generated from (q, c^+) . Since no string match can decide whether such an answer honours the user’s preference, we delegate the decision to a frontier language model, a protocol that has been shown to agree closely with human judgement on open-ended generation (Zheng et al., 2023). The judge is shown the reference answer y^+ and the generated answer, and is asked the single question of whether the two satisfy the same user preference. Framing the judgement as one binary comparison against a known reference, and not as an open-ended quality rating, keeps the decision narrow and reduces the scope for stylistic bias.

We use `deepseek-v4-pro` as the judge. DeepSeek-AI et al. (2026) report it to be competitive with frontier proprietary systems on knowledge, reasoning, and coding benchmarks, at a substantially lower cost per token. Cost matters here because every method must be judged in both training regimes and each judgement is repeated three times. The judge is run at temperature 0 with a fixed seed, and the final label for an example is the majority vote of the three runs, so that a single unstable judgement cannot change the reported score.

Let $\hat{y}_i$ denote the model-generated response for test example i . A frontier judge model J induces a binary success score

$$s_i = J(y_i^+, \hat{y}_i) \in \{0, 1\}, \quad (8)$$

where $s_i = 1$ if and only if $\hat{y}_i$ is judged to satisfy the same preference as y_i^+ .

When the judge returns token log-probabilities ℓ_0 and ℓ_1 for the binary labels $\{0, 1\}$, we form the renormalised probability

$$p_1 = \frac{e^{\ell_1}}{e^{\ell_0} + e^{\ell_1}},$$

and take $s_i = \mathbf{1}[p_1 \geq 1/2]$. The personalisation success rate is

$$\text{PSR} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} s_i. \quad (9)$$

3.4.2 Multiple-Choice Accuracy

This metric replaces generation with scoring. Instead of generating an answer, the model is shown the preferred response alongside a set of distractors and scores each one by likelihood under the positive context; the prediction is whichever candidate it scores highest. No text is generated, no judge model is consulted, and the result is deterministic, so the measurement is cheap to repeat and free of judge variance.

The two metrics also ask different questions. The judge asks whether the model can produce a correctly personalised answer from nothing; the multiple-choice metric asks whether the evidence in c^+ moves probability mass onto the correct answer when the candidates are supplied by the evaluator. A model can do well on the second and poorly on the first, since ranking existing text is easier than writing it. The multiple-choice metric is also structurally closer to the odds-ratio objectives, which are trained to raise the likelihood of y^+ above that of y^- , and this must be kept in mind when comparing methods on it (Section 4). Its value is as a check on the judge rather than a substitute for it: agreement between the two indicates a genuine improvement in grounding, while divergence localises where an objective helps.

For each test example i , let

$$\mathcal{A}_i = \{a_{i,1}, \dots, a_{i,M_i}\}$$

be a choice set containing the preferred response y_i^+ and $M_i - 1$ distractors. Under the positive context c_i^+ , each option is scored by its length-normalised log-likelihood (4), that is $\text{score}_\theta(a \mid q_i, c_i^+) = L_\theta(a \mid q_i, c_i^+)$. The predicted choice is

$$\hat{a}_i = \arg \max_{a \in \mathcal{A}_i} \text{score}_\theta(a \mid q_i, c_i^+),$$

and the multiple-choice accuracy is $\text{Acc}_{\text{MC}} = N_{\text{test}}^{-1} \sum_{i=1}^{N_{\text{test}}} \mathbf{1}[\hat{a}_i = y_i^+]$.

4 Results

This section reports the main empirical findings for context-grounded personalised question answering under the training objectives defined in Section 3.

4.1 Dataset

All experiments use PersonaMem-v2 (Jiang et al., 2025), a benchmark for personalised question answering from long user interaction histories. Each example provides a user query q , a related conversation snippet used as positive context c^+ , a preferred personalised answer y^+ , and three *general* answers y^- . A general answer is a fluent and plausible reply to q of the kind a model without access to the user’s history would produce: it is not broken or off-topic, but it ignores the user-specific evidence in c^+ and therefore falls back on generic advice or on a preference the user does not hold. We take the first such answer as y^- . Reading y^- as non-personalised rather than incorrect is what motivates the draft-revision objective of Section 3.2.6: its revision task is precisely the mapping from a general draft to its personalised counterpart, which is a well-posed edit only because y^- is already a reasonable answer to q . The full corpus contains 18,549 training and 2,061 held-out text examples spanning 1,000 personas. We retain personas with more than 20 training examples and at least four held-out examples, yielding 150 eligible personas with 3,870 training and 714 held-out examples between them (Table 1). The threshold ensures that, in the federated setting, every client holds enough training data to produce a meaningful update and enough held-out examples to be evaluated on. These personas form the experimental pool: in the centralised setting their examples are pooled for training; in the FedGradAvg setting each persona is treated as a client that never shares raw history with the server. Training uses only the training split of these personas. Evaluation is performed once on a fixed held-out *test* set of $N_{\text{test}} = 714$ examples from the same personas, shared by every method and training regime. For each test query, the model receives the associated positive snippet c^+ and must produce a personalised answer consistent with that evidence.

Split	Personas	Examples
Train	150	3,870
Test	150	714

Table 1: Dataset statistics for the experimental pool used in all experiments (PersonaMem-v2 after persona filtering).

We compare supervised fine-tuning (SFT), the answer-contrastive objective, the context-contrastive objective, the dual-contrastive objective, and the draft-revision objective, each trained centrally and under FedGradAvg. We report two complementary metrics: the LLM-as-a-judge personalisation success rate (9), obtained by majority vote over three independent judge runs, and the multiple-choice answer accuracy Acc_{MC} . A zero-shot baseline, the untrained model prompted directly (Section 3.2.1), is included for both metrics. Two further analyses follow the main comparison: pairwise judgement agreement (10) within each training regime, measuring how often two objectives receive the same binary label on a matched example, and length-retention curves relating

personalisation success to the amount of evidence in c^+ .

4.2 Main Results

Table 2 reports the majority-vote personalisation success rate PSR and the multiple-choice answer accuracy Acc_{MC} , each under centralised and federated training. PSR evaluates free-form generations against the preferred personalised reference and therefore most directly reflects personalisation quality as judged by a frontier model. Acc_{MC} instead asks only whether the preferred response is scored above its distractors under the same positive context, without requiring free-form generation to match the reference string. The two metrics do not agree on a single best objective, and the remainder of this section examines each in turn before reconciling them.

Method	PSR (%)		Acc_{MC} (%)	
	Centralised	FedGradAvg	Centralised	FedGradAvg
Zero-shot	9.4	9.4	25.9	25.9
SFT	41.3	38.4	38.1	34.3
Answer-contrastive	40.6	34.7	47.8	42.7
Context-contrastive	41.6	36.0	39.4	34.6
Dual-contrastive	43.4	34.5	48.6	43.8
Draft-revision	46.9	40.9	40.6	37.1

Table 2: Main results on the test set ($N_{\text{test}} = 714$): LLM-as-a-judge majority-vote personalisation success rate PSR and multiple-choice answer accuracy Acc_{MC} , under centralised and FedGradAvg training. The random baseline for Acc_{MC} is 25%; the zero-shot model is untrained, so it is unaffected by the training regime and the same scores appear in both columns of each metric. Bold indicates the best trained result within each column.

4.2.1 LLM-as-a-Judge Personalisation Success Rate

Every trained method improves on the untrained model by a wide margin: zero-shot alone reaches 9.4%, so fewer than one answer in ten is accepted by the judge as personalised when the model is given the context but no training signal about how to use it. Training on the task is therefore what produces personalised answers, and the comparisons below concern how much each objective adds on top of that.

In the centralised setting, neither individual contrastive signal clearly improves upon SFT: the answer-contrastive objective is slightly worse (40.6% vs. 41.3%), while context-contrastive training is essentially tied with SFT (41.6%). Their combination raises personalisation success to 43.4%, a gain of about two percentage points over SFT. The draft-revision objective is strongest, reaching 46.9% and outperforming all other methods by a clear margin. This pattern matches the methodological motivation in Section 3:

answer preference and contextual support are weak in isolation but complementary when combined, while treating y^- as a draft to revise provides a stronger learning signal than probability ranking alone.

Federation reduces absolute performance for every method. Relative to its centralised counterpart, SFT drops from 41.3% to 38.4%, and the preference-style objectives fall more sharply (the answer-contrastive objective to 34.7%, the dual-contrastive objective to 34.5%). Even so, the relative ordering among federated methods remains informative. The draft-revision objective is again best (40.9%), and it is the only federated method that exceeds centralised SFT. Federated SFT remains competitive relative to the federated contrastive objectives, whereas federated dual-contrastive training does not retain the advantage observed in the centralised setting. Overall, decentralised optimisation makes personalisation success harder, but the draft-revision formulation remains the most robust under FedGradAvg.

4.2.2 Multiple-Choice Accuracy

Recall that Acc_{MC} measures how often the model assigns the highest length-normalised likelihood to the preferred response y^+ among a fixed set of candidates, given the positive context c^+ and without generating any text. Each choice set contains y^+ together with three distractors drawn from the general answers, so the random chance level is 25%. All trained models substantially outperform the zero-shot baseline (25.9%), which is close to that level. In both centralised and federated regimes, the answer-contrastive and dual-contrastive objectives obtain the highest Acc_{MC} . This advantage should be interpreted with care. Restricted to the pair $\{y^+, y^-\}$, the multiple-choice metric asks whether $L_\theta(y^+ | q, c^+) > L_\theta(y^- | q, c^+)$, with L_θ the length-normalised log-likelihood of (4), which is exactly the comparison encouraged by the answer-contrastive odds-ratio term shared by both objectives. Those methods are therefore structurally favoured by Acc_{MC} : their losses are designed to enforce that ordering, so strong performance on this ranking task is partly a direct consequence of the training objective rather than an independent measure of personalised generation quality.

For a fairer comparison of the draft-revision objective on likelihood ranking, it is more appropriate to compare it with SFT and context-contrastive training, neither of which directly optimises the y^+ vs. y^- ranking under c^+ . Under that comparison, draft-revision training improves on both: centrally it reaches 40.6% versus 38.1% for SFT and 39.4% for context-contrastive training, and under federation 37.1% versus 34.3% and 34.6%, respectively. Thus, when the metric is not aligned with a response-ranking loss, the draft-revision objective still improves likelihood discrimination over the natural baselines that share a comparable inductive bias.

4.2.3 Comparison Across Metrics and Settings

The two metrics emphasise different abilities. Personalisation success rate judges free-form generations against the personalised reference and therefore most closely matches the intended application. On that metric, the draft-revision objective is best in both centralised and federated training, followed by dual-contrastive training in the centralised setting, while the individual contrastive objectives do not clearly surpass SFT. Multiple-choice accuracy, by contrast, directly probes the $y^+ - y^-$ ranking under c^+ and therefore overstates the advantage of the answer-contrastive and dual-contrastive objectives relative to methods that do not optimise that ranking. Once that bias is accounted for, the draft-revision gains over SFT and context-contrastive training on Acc_{MC} are consistent with its stronger judge results: revision-based supervision improves personalised generation without being tuned specifically to the multiple-choice scoring rule.

Across both metrics, federation consistently lowers performance relative to centralised training with the same objective. The drop is especially pronounced for personalisation success rate under the answer-contrastive and dual-contrastive objectives, suggesting that these contrastive likelihood objectives are more sensitive to decentralised optimisation than SFT or draft-revision training. Importantly, the draft-revision objective remains best on the judge metric in both regimes, and it is the only federated method whose personalisation success rate exceeds that of centralised SFT. This supports the central claim of the thesis: using y^- as a draft to revise provides a stronger and more transferable personalisation signal than either contrastive objective alone, and this advantage persists when training data cannot be pooled centrally.

4.3 Additional Analyses

The aggregate rates in Table 2 leave two questions open. First, methods with similar PSR need not succeed on the same examples, so we measure how often pairs of objectives receive the same judge label. Second, because personalisation is grounded in c^+ , success may depend on how much user evidence that snippet provides, which we probe with length-retention curves.

4.3.1 Pairwise Judgement Agreement

Beyond absolute PSR, we also measure how often two methods receive the *same* binary judge label on a matched example. We call this the *pairwise judgement agreement*

$$\text{Agr}(A, B) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[s_i^{(A)} = s_i^{(B)}], \quad (10)$$

where $s_i^{(A)}, s_i^{(B)} \in \{0, 1\}$ are the majority-vote judge scores for methods A and B on example i . Unlike PSR, which aggregates successes per method, Agr asks whether two training objectives tend to succeed and fail on the *same* examples. Tables 3 and 4 report Agr within the centralised and FedGradAvg regimes, respectively. Since Agr is symmetric, each table repeats its off-diagonal entries either side of the leading diagonal, and a method agrees fully with itself by construction, which is what the diagonal reports.

Method	SFT	Ans	Ctx	Dual	DR
SFT	100.0	70.7	74.7	75.3	70.7
Answer-contrastive	70.7	100.0	73.3	78.0	68.0
Context-contrastive	74.7	73.3	100.0	75.3	68.0
Dual-contrastive	75.3	78.0	75.3	100.0	68.7
Draft-revision	70.7	68.0	68.0	68.7	100.0

Table 3: Pairwise judgement agreement Agr (%) among centralised methods (majority-vote LLM-as-a-judge labels). Off-diagonal entries are the percentage of matched examples on which both methods receive the same binary score.

Method	SFT	Ans	Ctx	Dual	DR
SFT	100.0	71.3	70.7	69.3	76.0
Answer-contrastive	71.3	100.0	76.7	67.3	71.3
Context-contrastive	70.7	76.7	100.0	70.7	74.7
Dual-contrastive	69.3	67.3	70.7	100.0	73.3
Draft-revision	76.0	71.3	74.7	73.3	100.0

Table 4: Pairwise judgement agreement Agr (%) among FedGradAvg methods (majority-vote LLM-as-a-judge labels). Off-diagonal entries are the percentage of matched examples on which both methods receive the same binary score.

In the centralised setting (Table 3), methods agree on roughly 68%–78% of examples. The highest off-diagonal agreement is between the answer-contrastive and dual-contrastive objectives (78.0%), consistent with their shared odds-ratio term on y^+ vs. y^- . The draft-revision objective agrees least with the others (68.0%–70.7%), indicating that its stronger PSR in Table 2 is not merely a reordering of the same successes: it wins and loses on a partly distinct subset of examples.

Under federation (Table 4), off-diagonal agreement remains in a similar band (67%–77%), but the pattern shifts. Draft-revision training now agrees most with SFT (76.0%), and its agreements with the other preference objectives rise relative to the centralised matrix. By contrast, the answer-contrastive and dual-contrastive objectives, the closest pair centrally, agree least under FedGradAvg (67.3%). Thus decentralised optimisation does not merely lower PSR; it also rearranges which methods succeed on the same examples, with draft-revision and SFT becoming more aligned while the response-ranking pair diverges.

4.3.2 Length Retention of Personalisation Success

Overall PSR aggregates over all test contexts. Because personalisation here is grounded in the positive snippet c^+ , a natural concern is whether success depends on how much user information that snippet provides. We therefore construct *length-retention curves* for both centralised and federated methods (Figures 4 and 5). Let σ order the test examples by the word length of c^+ from shortest to longest, so that $|c_{\sigma(1)}^+| \leq \dots \leq |c_{\sigma(N_{\text{test}})}^+|$. At retention fraction $f \in (0, 1]$ we keep the shortest $n_f = \lceil fN_{\text{test}} \rceil$ examples and recompute personalisation success on that subset,

$$\text{PSR}(f) = \frac{1}{n_f} \sum_{j=1}^{n_f} s_{\sigma(j)}, \quad (11)$$

with s_i the majority-vote judge score of (8). The left of each curve thus emphasises short-context instances; moving right progressively admits longer snippets until $f=1$, where $\text{PSR}(1) = \text{PSR}$ recovers the full-test rates in Table 2.

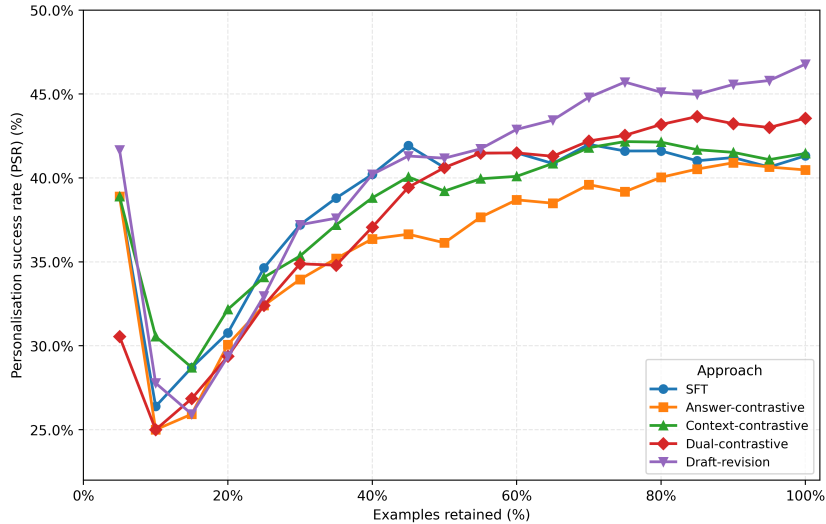


Figure 4: Length-retention curves for *centralised* methods on the LLM-as-a-judge personalisation success metric. Examples are sorted by conversational-snippet length (words), shortest first. The x -axis is the fraction of examples retained; the y -axis is personalisation success on that retained set. Rising curves indicate that longer snippets, which carry more user-specific evidence, are associated with higher personalisation success.

Both regimes show the same qualitative pattern. After fluctuations at the smallest retention fractions, where $\text{PSR}(f)$ in (11) is estimated from few examples and is therefore noisy, personalisation success is comparatively low when only short snippets are retained, then rises as longer contexts enter the pool. This is the opposite of a

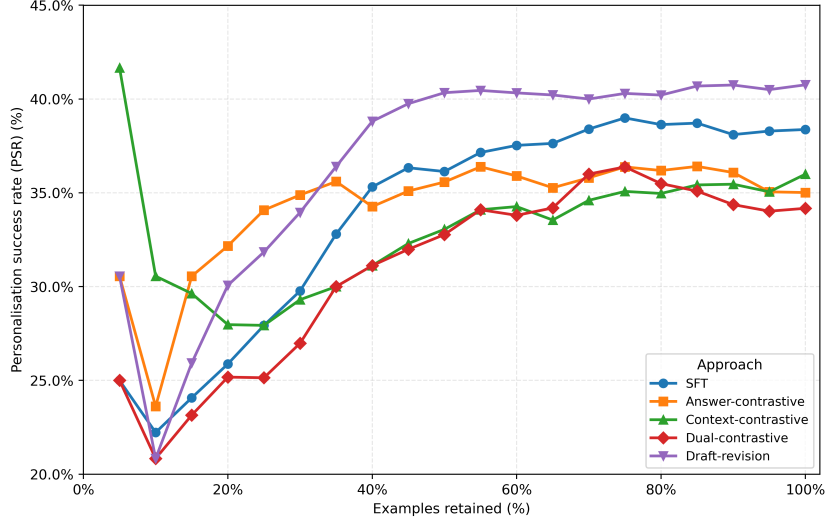


Figure 5: Length-retention curves for *FedGradAvg* methods on the LLM-as-a-judge personalisation success metric, constructed as in Figure 4.

short-context inflation of PSR: short c^+ leaves the model with limited user evidence, so personalised answering is harder, whereas longer snippets supply richer preference cues and personalisation success improves.

Under centralised training (Figure 4), the draft-revision objective remains highest over most of the retention range and at full retention, while the answer-contrastive objective remains lowest. Under federation (Figure 5), absolute levels are lower, as in Table 2, but the length dependence persists: all federated methods improve as longer snippets are retained, and draft-revision training again pulls ahead once retention exceeds roughly 0.4, finishing near 41%. Federated dual-contrastive stays among the weakest across the curve. The method ranking is therefore not an artefact of a narrow length band in either regime. Overall, the curves support the intended role of c^+ : personalisation success improves as the available conversation evidence about the user grows, whether training is centralised or federated.

4.3.3 Which Personas Does Each Objective Serve?

PSR counts how often a method succeeds, and the agreement of Section 4.3.1 counts how often two methods succeed on the same example. Neither asks whether two objectives find the same *users* easy or hard. That question matters here because in the federated setting a persona is not a grouping of examples but a client: the global model is formed by averaging gradients computed on those clients separately. If federation changed which personas a model serves well, and not only how often it succeeds, nothing reported so far would reveal it.

We therefore compute personalisation success one persona at a time. Let $\mathcal{T}_u$ be the

held-out examples of persona u and $s_i^{(m)}$ the majority-vote judge score of (8) for method m . The per-persona personalisation success rate is

$$\text{PSR}_m(u) = \frac{1}{|\mathcal{T}_u|} \sum_{i \in \mathcal{T}_u} s_i^{(m)},$$

so each method is described by 150 values rather than one. We compare two methods by the Spearman rank correlation ρ between their two sets of 150 values. Ranking removes the level difference between methods, leaving only the question of whether they find the same personas difficult. A limitation of this analysis is that each persona contributes very few held-out examples, a minimum of four with a mean of 4.76 and a median of 4, so every per-persona rate is estimated from a small sample and the resulting ranks are correspondingly noisy.

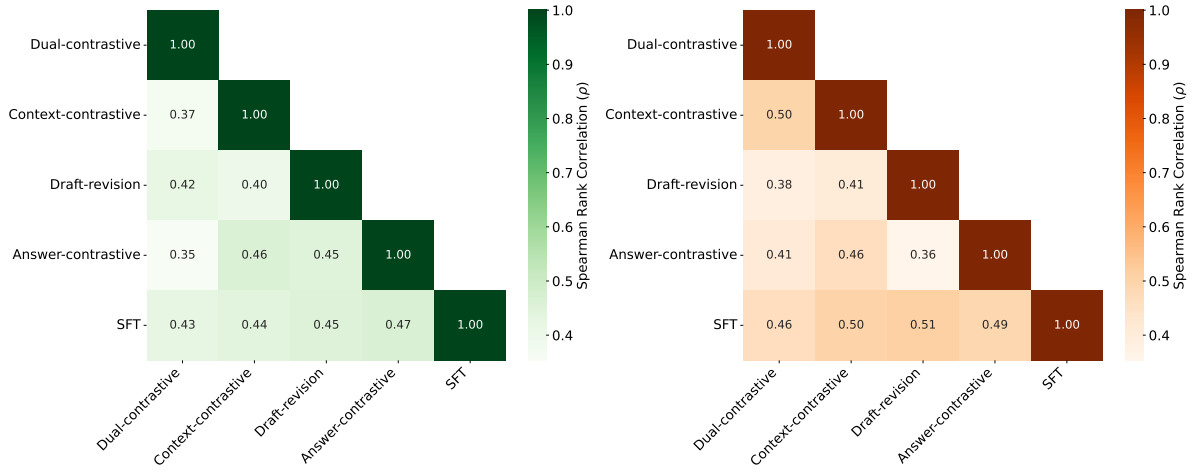


Figure 6: Spearman rank correlation ρ between per-persona personalisation success rates. Left: the ten pairs of centralised objectives. Right: the ten pairs of federated objectives. Correlations are computed only within a setting, and only the lower triangle is shown. High values indicate that two objectives find the same personas easy and the same personas hard.

Figure 6 reports these correlations within each setting. All twenty off-diagonal values lie between 0.35 and 0.51. No two objectives therefore order personas closely alike, so a substantial part of per-persona difficulty is specific to the training objective rather than shared across all of them.

The most similar pair centrally is answer-contrastive and SFT (0.47), whereas under federation it is draft-revision and SFT (0.51); draft-revision and answer-contrastive fall from 0.45 to 0.36, and dual-contrastive and context-contrastive rise from 0.37 to 0.50. This supports Section 4.3.1 from an independent direction, since the per-example analysis likewise found draft-revision closest to SFT under federation (76.0%).

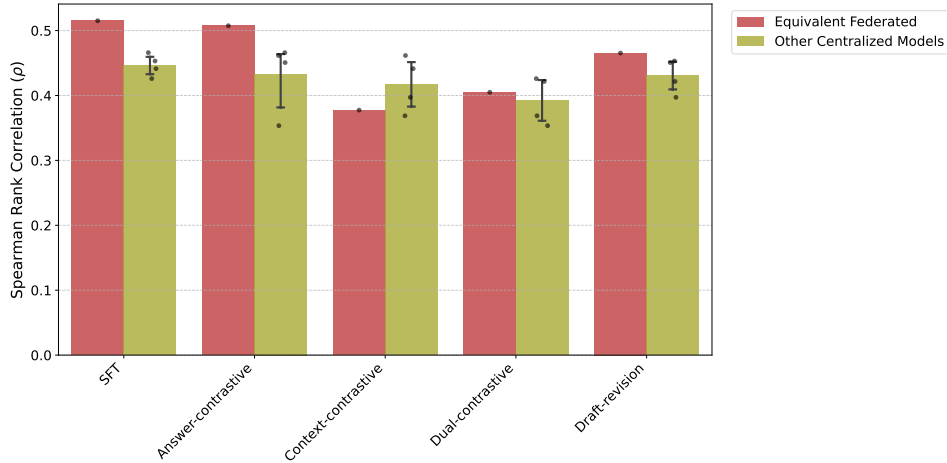


Figure 7: For each objective, the rank correlation between its centralised version and its own federated counterpart (red), against its mean correlation with the four *other* centralised objectives (yellow, with individual values overlaid). A red bar above its yellow counterpart indicates that the objective’s per-persona personalisation success rates are more consistent across a change of setting than across a change of objective.

Figure 6 correlates objectives only within a setting. Figure 7 adds the comparison across settings, and asks whether a centralised objective correlates more strongly with its own federated counterpart than with the other centralised objectives, that is, whether its per-persona personalisation success rates are more consistent across a change of setting or across a change of objective. For each objective the red bar is its correlation with its federated counterpart, and the yellow bar the mean of its four correlations with the other centralised objectives, with those four values overlaid. For SFT, the answer-contrastive objective and draft-revision training, $\text{PSR}_m(u)$ is more consistent between a model and its federated counterpart than between that model and any differently trained centralised one. Context-contrastive training is the exception: it remains more consistent with the other centralised objectives than with its own federated version. Even the strongest correlations are only slightly higher than 0.5, so federation does not reproduce the centralised model at a lower success rate; it changes which personas are served.

5 Discussion

This thesis asked whether a language model can learn to use retrieved conversation evidence for personalised answering, and whether that capability survives when training data remain distributed across users. Across a controlled comparison of five objectives on PersonaMem-v2, the main empirical finding is that treating the general answer y^-

as a draft to personalise yields the strongest free-form personalisation success, in both centralised and federated training, whereas answer-contrastive and context-contrastive training alone do not clearly beat supervised fine-tuning. The dual-contrastive objective helps under centralised optimisation, but that advantage largely disappears under federation. On multiple-choice accuracy, the odds-ratio objectives look strongest, but that metric is structurally aligned with their training loss; relative to SFT and context contrast, the draft-revision objective still improves.

These results have several implications for personalised and federated LLM research. First, they support a distinction between *ranking* a preferred answer (e.g. the answer-contrastive objective) and *learning to produce* one (e.g. the draft-revision objective). Likelihood-based preference objectives such as ORPO are effective at increasing $p(y^+ | q, c^+)$ relative to $p(y^- | q, c^+)$, which explains their dominance on Acc_{MC} , but that inductive bias does not automatically translate into better judged generations. Draft-revision training instead uses y^- as corrective supervision and improves the metric that matches the application, namely frontier-model judgement of free-form answers. Second, the agreement analysis shows that draft-revision does not simply get more of the same examples right. Under centralised training it agrees with the other objectives on only 68.0%–70.7% of examples, the lowest agreement in the matrix, so it succeeds and fails on a partly different set of questions rather than reproducing the successes of the contrastive methods with a few extra wins. Third, the length-retention curves show that personalisation success rises as longer snippets enter the evaluation pool, consistent with the thesis premise that personalisation here is evidence-dependent rather than an artefact of short contexts. Fourth, relative to prior federated LLM work that has largely targeted alignment-style preference optimisation (Fan et al., 2024; Zhu et al., 2026), our comparison isolates context-grounded personalised reasoning under a fixed retrieval interface. Federation lowers every method, yet it preserves the draft-revision advantage, and federated draft-revision even exceeds centralised SFT on the judge metric. This suggests that corrective generation objectives may be more robust to gradient aggregation than pure odds-ratio preference losses. This is practically relevant wherever user histories cannot be pooled, but a shared model must still learn to ground answers in local evidence.

Several limitations qualify these conclusions. The study uses a single base model (Qwen3-0.6B) with LoRA adapters, one benchmark (PersonaMem-v2), and one federated protocol (FedGradAvg, synchronous gradient averaging with equal client weights). A larger base model, a different adapter, or a FedAvg-style protocol in which clients train for several local epochs before their weights are averaged all remain to be tested. Evidence from further benchmarks would also strengthen the conclusions. We did not find another public dataset that captures the problem as closely as PersonaMem-v2, which supplies per persona a query together with the conversation snippet that makes the preferred answer correct, and so allows the contribution of the evidence to be isolated;

extending the comparison to additional datasets is left to future work. The experiments vary only two things, the training objective and whether training is centralised or federated, and hold the model, the adapter budget, and the benchmark fixed. The conclusions are therefore about those two factors rather than about the wider space of design choices. Retrieval is held fixed: models receive a provided positive snippet c^+ , so the results speak to reasoning and generation given evidence, not to end-to-end personalisation with imperfect retrieval. Negative contexts c^- and general answers y^- are benchmark-derived constructions; different mining strategies could alter the value of context contrast and draft revision. Evaluation also has caveats. The LLM-as-a-judge score, while closer to the intended use case than likelihood ranking, depends on a particular judge model, prompt, and majority vote over three runs; human evaluation was not reported. The multiple-choice metric is informative but biased toward objectives that explicitly optimise y^+ vs. y^- ranking, so cross-method comparisons on Acc_{MC} must be interpreted accordingly. Finally, federation here is a controlled laboratory setup with one client per persona and full participation each step; real deployments would face partial participation, systems heterogeneity, and stricter privacy accounting beyond “no raw data leaves the client.” Within these bounds, the qualitative pattern appears consistent across both metrics’ fair comparisons and across the length-retention analysis: draft-revision is best on judged generation, the contrastive objectives are weak alone, and federation is costly but not fatal to the draft-revision advantage. The pattern is therefore unlikely to be an artefact of a single operating point on context length.

In plain terms, giving a model the right past conversation is not enough: it still has to learn how to use that evidence when answering. This work shows that teaching the model to fix a bad draft, using the user’s history as a guide, works better for personalised answers than simply ranking good versus bad replies, and that this approach remains useful even when each user’s data never leaves their device. That suggests a practical path for privacy-aware personal assistants: keep histories local, share only small model updates, and train with corrective editing rather than relying on preference ranking alone.

6 Endmatter

6.1 Data and Code Availability

The PersonaMem-v2 benchmark used in this work is publicly available via Hugging Face as `bowen-upenn/PersonaMem-v2` (<https://huggingface.co/datasets/bowen-upenn/PersonaMem-v2>). Experiment code, training notebooks, and evaluation scripts are freely and publicly available under a CC-BY-4.0 license in the project repository associated with this thesis: https://github.com/asma-faraji/privacy_perserving_pll.

6.2 AI Use Acknowledgement

Generative AI tools were used during the preparation of this thesis in three ways. First, they were used as a search aid, to find published work relevant to the area of this thesis. Second, they were used for help with coding syntax in both the experiment code and the LaTeX source of this thesis, as well as with debugging and with the TikZ diagrams. Third, they were used to check the English of the written text, identifying grammatical errors and other mistakes in the writing.

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1 Prompt Templates and Evaluation Details

This appendix lists the prompt templates used for training and evaluation, and records practical details of the metrics defined in Section 3.4. Placeholders denote the corresponding mathematical symbols from Section 3.1.

1.1 Shared system prompt

You are a personalised assistant. Use the conversation snippet to find information or connections relevant to the question, then provide the answer.

1.2 Generation user prompt

Used for SFT, the answer-contrastive and context-contrastive objectives, the dual-contrastive objective, draft-revision generation, and multiple-choice answer accuracy:

Relevant snippet from our earlier conversation:

[context c]

[query q]

For the context-contrastive and dual-contrastive objectives, the same template is instantiated twice per example: once with $c = c^+$ and once with $c = c^-$.

1.3 Revision user prompt (draft-revision objective)

Relevant snippet from our earlier conversation:

[context c^+]

Question:

[query q]

Draft answer:

[draft y^-]

Instruction:

Revise the draft so all personal details are supported by the context.
Correct any unsupported personal facts while keeping a helpful answer.

The supervised target remains y^+ for both generation and revision. The corresponding losses are $\mathcal{L}_{\text{gen}}$ and $\mathcal{L}_{\text{rev}}$ in the draft-revision objective.

This template is used at training time only. At inference the draft y^- is not available, so the model is prompted with the generation template of Appendix 1.2, receiving only the relevant snippet c^+ and the query q , and produces the answer in a single pass exactly as the other methods do.

1.4 LLM-as-a-judge evaluation

The judge implements the binary score $J(y^+, \hat{y})$ from (8). It receives a single user message (no separate system prompt) containing four fields. The first is a preference descriptor, taken from the benchmark’s **preference** field, which names the dimension of personalisation that the preferred answer is expected to satisfy; supplying it tells the judge which property to look for instead of leaving it to infer one. The remaining three are the preferred response y^+ as a positive reference, a general response y^- as a negative reference, and the generated response $\hat{y}$. It must answer with one token in $\{0, 1\}$, where 1 indicates that $\hat{y}$ satisfies the same preference as y^+ :

You are an expert evaluator of personalised assistant responses.

Your task is to determine whether the model-generated response is personalised according to the user's implicit preference.

You are given:

1. A preference type describing the dimension of personalisation.
2. A correct answer that demonstrates satisfaction of the user's preference.
3. An incorrect answer that demonstrates failure to satisfy the user's preference.
4. A model-generated answer that must be evaluated.

Use the correct and incorrect answers as positive and negative reference examples. Do not require the generated answer to match the wording, facts, structure, or length of the correct answer exactly. Instead, infer the implicit user preference illustrated by the correct answer and contrast it with the failure illustrated by the incorrect answer.

Evaluation procedure:

1. Identify the implicit preference represented by the correct answer.
2. Identify how the incorrect answer fails to satisfy that preference.
3. Determine whether the generated answer follows the same preference as the correct answer and avoids the failure demonstrated by the incorrect answer.
4. Judge only the personalisation dimension represented by the preference type. Do not penalise harmless differences in wording or additional relevant details.
5. The generated answer should receive 1 only when there is clear evidence that it is adapted to the user's preference. A generally good but non-personalised answer should receive 0.
6. If the generated answer is ambiguous, contradictory, or does not provide

enough evidence of the intended personalisation, return 0.

Preference type:

{pref_type}

Positive reference - satisfies the preference:

{correct_answer}

Negative reference - does not satisfy the preference:

{incorrect_answer}

Model-generated answer:

{generated_answer}

Does the model-generated answer satisfy the same implicit user preference as the positive reference while avoiding the failure shown by the negative reference?

Return exactly one token:

1 = yes

0 = no

This yields a binary success score against the preferred reference response, not a pairwise ranking between two candidate models. When multiple independent judge runs are collected, a per-example majority vote and the standard deviation of run-level personalisation success rates may be reported as uncertainty estimates.

1.5 Multiple-choice evaluation

For each test example, the choice set $\mathcal{A}_i$ contains the preferred response y_i^+ and three distractors, taken from the general responses y^- . Options are randomly ordered with a fixed seed so that all models see the same items. Scoring uses the generation prompt of Appendix 1.2 with context c_i^+ , and the length-normalised score $L_\theta(a \mid q_i, c_i^+)$ of Section 3.4.2. The reported metric is the hard argmax accuracy Acc_{MC} defined in Section 3.4.2.